

How to use this interactive PDF: Click **Show / Hide Answer** under any question to reveal or hide its answer. This requires **Adobe Acrobat Reader** (free). In other viewers the buttons may not work — use your viewer's **Layers** panel, or the **Reveal All** button below.

Reveal All

Hide All

CCNA Practice Question Bank

100+ Exam-Style Questions with Answers & Full Explanations

How to use it: Try each question BEFORE reading the answer. The explanation tells you why the right answer is right AND why the traps are wrong — that's what builds real understanding. Aim to consistently score **85%+** before your exam. Questions are grouped by the 6 official exam domains.

Domain 1 — Network Fundamentals

Q1. Which device operates primarily at Layer 2 of the OSI model?

A. Router B. Switch C. Firewall D. Load balancer

Show / Hide Answer

Q2. What is the correct order of TCP's three-way handshake?

A. ACK, SYN, SYN-ACK B. SYN, ACK, SYN-ACK C. SYN, SYN-ACK, ACK D. SYN-ACK, SYN, ACK

Show / Hide Answer

Q3. Which protocol is connectionless and does NOT guarantee delivery?

A. TCP B. UDP C. HTTP D. FTP

Show / Hide Answer

Q4. What is the data unit at Layer 3 called?

A. Frame B. Segment C. Packet D. Bit

Show / Hide Answer

Q5. Which port number does HTTPS use?

A. 80 B. 21 C. 443 D. 22

Show / Hide Answer

Q6. A host has IP 169.254.10.5. What most likely happened?

A. It got an address from DHCP B. It was assigned a public IP C. It could not reach a DHCP server (APIPA) D. It is using a loopback

Show / Hide Answer

Q7. Which cable type connects two switches together (without Auto-MDIX)?

A. Straight-through B. Crossover C. Rollover D. Fiber only

Show / Hide Answer

Q8. How many bits are in an IPv6 address?

A. 32 B. 48 C. 64 D. 128

Show / Hide Answer

Q9. Which of the following is a private IPv4 address?

A. 8.8.8.8 B. 172.32.0.1 C. 192.168.5.20 D. 200.1.1.1

Show / Hide Answer

Q10. What does the Transport layer use to identify which application traffic belongs to?

A. MAC addresses B. IP addresses C. Port numbers D. VLAN tags

Show / Hide Answer

Q11. Which IPv6 address type replaces the IPv4 broadcast?

A. Anycast B. Multicast C. Unicast D. Loopback

Show / Hide Answer

Q12. Convert binary 10101100 to decimal.

A. 172 B. 168 C. 160 D. 174

Show / Hide Answer

Q13. Which layer of the TCP/IP model maps to OSI Layers 5, 6, and 7?

A. Transport B. Internet C. Application D. Network Access

Show / Hide Answer

Q14. What is the broadcast MAC address?

A. 00:00:00:00:00:00 B. FF:FF:FF:FF:FF:FF C. 01:00:5E:00:00:01 D. FE80::1

Show / Hide Answer

Q15. A fiber run must go 10 km between buildings. Which media is appropriate?

A. Cat6 copper B. Multimode fiber C. Single-mode fiber D. Coaxial

Show / Hide Answer

Domain 2 — Network Access (Switching, VLANs, Wireless)

Q16. What does a switch do when it receives a frame with an unknown destination MAC?

A. Drops it B. Floods it out all ports except the incoming one C. Sends it to the default gateway D. Buffers it forever

Show / Hide Answer

Q17. Which command puts switch port Fa0/1 into VLAN 10?

A. `vlan 10` B. `switchport access vlan 10` C. `switchport trunk vlan 10` D. `ip vlan 10`

Show / Hide Answer

Q18. Which standard is used to tag frames on a trunk link?

A. 802.11 B. 802.1Q C. 802.3 D. 802.1X

Show / Hide Answer

Q19. On an 802.1Q trunk, which VLAN's traffic is sent untagged by default?

A. VLAN 0 B. VLAN 1 (native) C. VLAN 1000 D. All VLANs

Show / Hide Answer

Q20. What is the purpose of Spanning Tree Protocol?

A. Speed up routing B. Prevent Layer 2 loops C. Assign IP addresses D. Encrypt traffic

Show / Hide Answer

Q21. In STP, how is the root bridge chosen?

A. Highest IP address B. Lowest bridge ID (priority + MAC) C. Fastest port D. Most connections

Show / Hide Answer

Q22. Which feature lets a switch port skip STP's listening/learning states for end devices?

A. BPDU Guard B. PortFast C. EtherChannel D. Root Guard

Show / Hide Answer

Q23. What happens by default when a port-security violation occurs?

A. Port logs only B. Port drops traffic silently C. Port shuts down (err-disabled) D. Port reboots

Show / Hide Answer

Q24. Which LACP mode combination will successfully form an EtherChannel?

A. passive/passive B. active/passive C. auto/auto D. on/active

Show / Hide Answer

Q25. Which two channels do NOT overlap on the 2.4 GHz band (pick the standard set)?

A. 1, 2, 3 B. 1, 5, 9 C. 1, 6, 11 D. 2, 7, 12

Show / Hide Answer

Q26. Which wireless security standard is the most secure?

A. WEP B. WPA C. WPA2 D. WPA3

Show / Hide Answer

Q27. In a lightweight wireless architecture, what protocol do APs use to talk to the WLC?

A. CDP B. CAPWAP C. LLDP D. GRE

Show / Hide Answer

Q28. Router-on-a-stick uses what to route between VLANs on one physical link?

A. Loopbacks B. Subinterfaces C. SVIs D. Trunk ACLs

Show / Hide Answer

Q29. Which command shows which VLAN each access port belongs to?

A. `show mac address-table` B. `show vlan brief` C. `show ip route` D. `show cdp neighbors`

Show / Hide Answer

Q30. What is the default native VLAN, and why change it?

A. VLAN 0; performance B. VLAN 1; security (avoid VLAN hopping) C. VLAN 99; redundancy D. VLAN 4094; naming

Show / Hide Answer

Domain 3 — IP Connectivity (Routing & OSPF)

Q31. In the routing table, what does the letter **R** mean?

A. A static route B. A connected route C. An OSPF-learned route D. An offline route

Show / Hide Answer

Q32. A router has two routes to the same network: one via OSPF, one static. Which is used?

A. OSPF (AD 110) B. Static (AD 1) C. Both load-balance D. Neither

Show / Hide Answer

Q33. Which administrative distance does a directly connected route have?

A. 0 B. 1 C. 90 D. 110

Show / Hide Answer

Q34. What does a router do with a packet if no matching route exists and there is no default route?

A. Floods it B. Sends to gateway of last resort C. Drops it D. Buffers it

Show / Hide Answer

Q35. Which command creates a default route toward next hop 203.0.113.1?

A. `ip route 0.0.0.0 0.0.0.0 203.0.113.1` B. `ip default 203.0.113.1` C. `ip route default 203.0.113.1`
D. `default-gateway 203.0.113.1`

Show / Hide Answer

Q36. When two routes to the same destination have different prefix lengths, the router chooses the one that is:

A. Learned first B. Most specific (longest prefix) C. From the lowest AD D. Load-balanced

Show / Hide Answer

Q37. What metric does OSPF use to choose the best path?

A. Hop count B. Bandwidth-based cost C. Delay D. Ticks

Show / Hide Answer

Q38. Two OSPF routers won't become neighbors. Which mismatch could be the cause?

A. Different hostnames B. Different interface descriptions C. Different area or subnet D. Different clock time

Show / Hide Answer

Q39. In OSPF, what must the backbone area always be?

A. Area 1 B. Area 10 C. Area 0 D. Area 255

Show / Hide Answer

Q40. What is a wildcard mask of `0.0.0.255` equivalent to matching?

A. A single host B. A whole /24 network C. Everything D. A /16 network

Show / Hide Answer

Q41. Which command correctly advertises 10.1.1.0/24 into OSPF area 0?

A. `network 10.1.1.0 255.255.255.0 area 0` B. `network 10.1.1.0 0.0.0.255 area 0` C. `network 10.1.1.0 area 0` D. `advertise 10.1.1.0/24`

Show / Hide Answer

Q42. How does OSPF pick its Router ID if not set manually?

A. Lowest interface IP B. Highest loopback, else highest active physical IP C. The MAC address D. Randomly

Show / Hide Answer

Q43. A floating static route is created by:

A. Lowering its metric B. Giving it a higher administrative distance than the primary C. Adding a wildcard mask D. Using a /32

Show / Hide Answer

Q44. Which routing protocol type builds a full map of the network and runs SPF?

A. Distance vector B. Link state C. Path vector D. Static

Show / Hide Answer

Q45. What is the OSPF cost of a 100 Mbps link using the default reference bandwidth?

A. 1 B. 10 C. 19 D. 100

Show / Hide Answer

Domain 4 — IP Services (DHCP, DNS, NAT, NTP, SNMP)

Q46. What is the correct order of the DHCP process?

A. Offer, Discover, Request, Ack B. Discover, Offer, Request, Ack C. Request, Offer, Discover, Ack D. Discover, Request, Offer, Ack

Show / Hide Answer

Q47. Which service translates domain names into IP addresses?

A. DHCP B. DNS C. NAT D. NTP

Show / Hide Answer

Q48. Your home router lets 20 devices share one public IP. What technology is this?

A. Static NAT B. Dynamic NAT C. PAT (NAT overload) D. Proxy ARP

Show / Hide Answer

Q49. DHCP clients are on a different subnet than the DHCP server. What must you configure on the router?

A. `ip default-gateway` B. `ip helper-address <server>` C. `ip nat inside` D. `ip route`

Show / Hide Answer

Q50. Which port does DNS primarily use?

A. 53 B. 67 C. 80 D. 123

Show / Hide Answer

Q51. Which DNS record maps a name to an IPv6 address?

A. A B. AAAA C. CNAME D. MX

Show / Hide Answer

Q52. Why is NTP important on network devices?

A. Faster routing B. Accurate timestamps for logs and certificates C. More bandwidth D. Encryption

Show / Hide Answer

Q53. Which SNMP version adds encryption and authentication?

A. SNMPv1 B. SNMPv2c C. SNMPv3 D. SNMPv2

Show / Hide Answer

Q54. In NAT terminology, the private address of a host BEFORE translation is the:

A. Inside local B. Inside global C. Outside local D. Outside global

Show / Hide Answer

Q55. A syslog message at severity level 0 means:

A. Debugging B. Informational C. Emergency (system unusable) D. Warning

Show / Hide Answer

Domain 5 — Security Fundamentals

Q56. Which part of the CIA triad ensures data isn't secretly altered?

A. Confidentiality B. Integrity C. Availability D. Authentication

Show / Hide Answer

Q57. Which attack fills a switch's MAC table to force flooding?

A. VLAN hopping B. MAC flooding C. DHCP starvation D. ARP spoofing

Show / Hide Answer

Q58. Which feature stops a rogue DHCP server on an untrusted port?

A. BPDU Guard B. DHCP snooping C. Port-security D. DAI

Show / Hide Answer

Q59. Which protocol should replace Telnet for remote device management?

A. FTP B. HTTP C. SSH D. SNMPv1

Show / Hide Answer

Q60. In AAA, what does the second "A" (Authorization) determine?

A. Who you are B. What you're allowed to do C. What you did D. Where you are

Show / Hide Answer

Q61. Which AAA protocol encrypts the entire packet and uses TCP?

A. RADIUS B. TACACS+ C. Kerberos D. LDAP

Show / Hide Answer

Q62. In 802.1X, what role does the switch play?

A. Supplicant B. Authenticator C. Authentication server D. Client

Show / Hide Answer

Q63. Which command encrypts plain-text passwords in the running config?

A. `enable secret` B. `service password-encryption` C. `crypto key generate rsa` D. `password-encrypt on`

Show / Hide Answer

Q64. Which defense protects against ARP spoofing (man-in-the-middle)?

A. Port-security B. Dynamic ARP Inspection (DAI) C. BPDU Guard D. STP

Show / Hide Answer

Q65. A site-to-site VPN between two offices typically uses:

A. SSL only B. IPsec C. Telnet D. GRE only

Show / Hide Answer

Domain 5b — Access Control Lists (ACLs)

Q66. What is at the end of every ACL, even if not typed?

A. permit any B. an implicit deny all C. a log entry D. nothing

Show / Hide Answer

Q67. A standard ACL filters based on what?

A. Source and destination IP B. Source IP only C. Port numbers D. MAC addresses

Show / Hide Answer

Q68. Where should a standard ACL be placed?

A. Close to the source B. Close to the destination C. On every interface D. On the trunk

Show / Hide Answer

Q69. In ACL processing, which rule wins?

A. The last match B. The first match C. The most specific D. The lowest number

Show / Hide Answer

Q70. Which statement permits only host 192.168.1.10?

A. `permit 192.168.1.10 0.0.0.255` B. `permit host 192.168.1.10` C. `permit any` D. `permit 192.168.1.0 0.0.0.0`

Show / Hide Answer

Domain 6 — Automation & Programmability

Q71. In SDN, which plane makes forwarding decisions?

A. Data plane B. Control plane C. Management plane D. Physical plane

Show / Hide Answer

Q72. A northbound API on a controller communicates with:

A. Switches B. Routers C. Applications/scripts D. The data plane

Show / Hide Answer

Q73. Which HTTP method is used to RETRIEVE data from a REST API?

A. POST B. GET C. PUT D. DELETE

Show / Hide Answer

Q74. Which data format uses curly braces and key/value pairs and is most common in REST APIs?

A. XML B. YAML C. JSON D. CSV

Show / Hide Answer

Q75. Which automation tool is agentless and uses YAML playbooks?

A. Puppet B. Chef C. Ansible D. SaltStack

Show / Hide Answer

Q76. What is the main benefit of network automation?

A. More cables B. Consistent, fast, error-free configuration at scale C. Slower changes D. Replacing all routers

Show / Hide Answer

Q77. Cisco DNA Center is an example of:

A. A router B. An SDN controller C. A firewall D. A switch OS

Show / Hide Answer

Domain 7 — Configuration & Troubleshooting Scenarios

Q78. After configuring an interface, `show ip interface brief` shows "administratively down." What fixes it?

A. `no shutdown` B. `ip address` C. Reboot D. `enable`

Show / Hide Answer

Q79. You can ping 8.8.8.8 but not google.com. What's broken?

A. Routing B. DNS C. NAT D. The cable

Show / Hide Answer

Q80. Two switches won't form a trunk. Which is a likely cause?

A. Different hostnames B. Native VLAN or mode mismatch C. Different IOS versions D. Different port numbers

Show / Hide Answer

Q81. A PC gets 169.254.x.x. What should you check first?

A. DNS server B. DHCP reachability (server/relay/VLAN) C. The default route D. STP

Show / Hide Answer

Q82. A link is slow with many errors. One side is full-duplex, the other half. This is a:

A. Speed mismatch B. Duplex mismatch C. VLAN mismatch D. MTU mismatch

Show / Hide Answer

Q83. Which command shows the path packets take and where they stop?

A. `ping` B. `tracert` C. `show ip route` D. `show cdp neighbors`

Show / Hide Answer

Q84. A port-security-enabled port is err-disabled after a violation. How do you recover it?

A. `no shutdown` only B. `shutdown` then `no shutdown` C. Reload D. Delete the VLAN

Show / Hide Answer

Q85. Which command shows OSPF neighbors and their states?

A. `show ip ospf neighbor` B. `show ip route` C. `show running-config` D. `show ip protocols`

Show / Hide Answer

Q86. A host's gateway is 192.168.1.65 /26, and the host is 192.168.1.130 /26. Why no internet?

A. Wrong DNS B. Host and gateway are in different subnets C. Bad cable D. STP blocking

Show / Hide Answer

Q87. Which command saves the running configuration so it survives a reboot?

A. `write erase` B. `copy running-config startup-config` C. `reload` D. `show startup-config`

Show / Hide Answer

Q88. `show cdp neighbors detail` is most useful for:

A. Seeing routing tables B. Discovering a neighbor's IP and platform C. Checking VLANs D. Viewing ACLs

Show / Hide Answer

Q89. A switch's MAC address table is empty for a device that's clearly connected and powered. What's a likely reason?

A. The device hasn't sent any frames yet B. STP is off C. Wrong IOS D. NAT is disabled

Show / Hide Answer

Q90. Which show command quickly confirms an interface is "up/up" with its IP?

A. `show vlan brief` B. `show ip interface brief` C. `show mac address-table` D. `show version`

Show / Hide Answer

Domain 8 — More Fundamentals & Mixed Review

Q91. How many usable hosts are in a /27 subnet?

A. 32 B. 30 C. 62 D. 14

Show / Hide Answer

Q92. Which address is the broadcast for 192.168.1.0/26's second subnet?

A. 192.168.1.63 B. 192.168.1.127 C. 192.168.1.128 D. 192.168.1.191

Show / Hide Answer

Q93. Which IPv6 address is a link-local address?

A. 2001:db8::1 B. FE80::1 C. ::1 D. FF02::1

Show / Hide Answer

Q94. What does :: mean in an IPv6 address?

A. End of address B. One or more groups of all zeros (used once) C. A separator only D. A loopback

Show / Hide Answer

Q95. Which protocol automatically lets an IPv6 host build its own address from the router's prefix?

A. DHCPv4 B. SLAAC C. ARP D. NAT

Show / Hide Answer

Q96. Which of these is a Layer 1 problem?

A. Wrong VLAN B. Unplugged/broken cable C. Missing default route D. ACL blocking

Show / Hide Answer

Q97. What is the purpose of a default gateway on a host?

A. Resolve names B. Reach devices on other networks C. Assign IPs D. Encrypt traffic

Show / Hide Answer

Q98. Which command tests basic connectivity to an IP?

A. ping B. show run C. configure terminal D. copy

Show / Hide Answer

Q99. A trunk carries VLANs 10 and 20, but VLAN 20 traffic isn't passing while VLAN 10 works. What's a likely cause?

A. Wrong hostname B. VLAN 20 not allowed on the trunk C. Duplex mismatch D. Wrong DNS

Show / Hide Answer

Q100. Which two are valid reasons to use VLANs? (Choose the best single answer.)

A. Faster CPUs B. Segment broadcast domains and improve security C. Replace routers D. Increase cable length

Show / Hide Answer

Q101. Which is TRUE about a /30 subnet?

A. It has 4 usable hosts B. It has 2 usable hosts, ideal for router links C. It's used for large LANs D. It has no broadcast

Show / Hide Answer

Q102. In the frame, which field detects transmission errors?

A. Preamble B. Source MAC C. FCS (Frame Check Sequence) D. Type

Show / Hide Answer

Q103. Which command sets a device to only allow SSH (not Telnet) for remote logins?

A. `transport input ssh` B. `transport input all` C. `no ip telnet` D. `login local`

Show / Hide Answer

Q104. What best describes "longest prefix match"?

A. Trusting the lowest AD B. Choosing the most specific matching route C. Load balancing D. Picking the oldest route

Show / Hide Answer

Q105. You need 6 subnets from a /24, each with room for ~25 hosts. Which mask fits BOTH needs?

A. /26 B. /27 C. /28 D. /25

Show / Hide Answer

How to Use These Results

- **Score yourself by domain.** If you miss several in one domain (say OSPF or ACLs), go back to that chapter in the main guide and re-drill.
- **Re-take after a few days.** Spacing your practice helps memory stick.
- **Target 85%+ consistently** before booking the exam.
- **Pair with labs.** Reading answers isn't enough — configure these things in Packet Tracer to make them permanent.

Good luck — you're building real skills now!

— End of Practice Question Bank —